

Self-Activating Hybrid Biological–Cybernetic Organisms Revisited

A Hybrid-Systems Model of Activation, Interface Limits, and Safety

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Abstract

We model a hybrid biological–cybernetic organism (HBCO) as a hybrid dynamical system with a biological state, a computational state, a bounded energy store, and two operating modes (dormant and active). Using standard results on hybrid systems and information theory, we show (i) that autonomous activation cycles exist iff harvested power lies strictly between dormant and active consumption, with bounded dwell times and a closed-form duty cycle, (ii) that information reaching the computational side, and any partition-based integration measure across the bio-cyber boundary, is bounded by the interface channel capacity, and (iii) that these results yield design requirements for containment that do not depend on software. Claims about life, evolution, and consciousness are stated as conditional or left open.

The paper ends with “The End”

1 Model

Following the cybernetic tradition [1, 2], an HBCO has state $x = (b, c)$, where $b \in \mathbb{R}^m$ is the biological state and $c \in \mathbb{R}^k$ the computational state. Let $E \in [0, E_{\max}]$ be a stored-energy variable and $q \in \{0, 1\}$ the mode (dormant or active). The hybrid system [3] is

$$\dot{x} = f_q(x) + \eta(t), \quad (1)$$

$$\dot{E} = P_h(t) - P_q, \quad P_0 = P_d, \quad P_1 = P_a, \quad (2)$$

where f_q are the mode-dependent dynamics (which include the interface coupling between b and c), η is bounded noise, P_h is harvested power, and P_q is consumption in mode q . Modes switch with hysteresis:

$$q : 0 \rightarrow 1 \quad \text{when } E \geq E_{\text{on}}, \quad (3)$$

$$q : 1 \rightarrow 0 \quad \text{when } E \leq E_{\text{off}}, \quad (4)$$

with $0 \leq E_{\text{off}} < E_{\text{on}} \leq E_{\max}$. The original paper’s equation $\dot{x} = \varphi(I(b, c)) + \eta$ is the special case with no intrinsic dynamics and no modes, in which self-activation cannot be represented. Let $\Delta E = E_{\text{on}} - E_{\text{off}}$.

2 Self-activation

Proposition 1 (Activation cycle). *Suppose $P_h(t) \in [P_h^-, P_h^+]$ for all t .*

1. *If $P_h^+ \leq P_d$, the system never activates from $E < E_{\text{on}}$.*

2. If $P_h^- \geq P_a$, once active it never returns to the dormant mode.
3. If $P_d < P_h^-$ and $P_h^+ < P_a$, the system cycles forever with no external trigger. Dormant dwell times lie in $[\Delta E/(P_h^+ - P_d), \Delta E/(P_h^- - P_d)]$ and active dwell times lie in $[\Delta E/(P_a - P_h^-), \Delta E/(P_a - P_h^+)]$.
4. For constant P_h in case 3, the duty cycle (fraction of time active) is

$$d = \frac{P_h - P_d}{P_a - P_d}. \quad (5)$$

Proof. In the dormant mode $\dot{E} = P_h - P_d$, which is bounded below by $P_h^- - P_d > 0$ and above by $P_h^+ - P_d$. Moving from E_{off} to E_{on} therefore takes a time within the stated interval, and similarly in the active mode, where $\dot{E} = P_h - P_a < 0$. For constant P_h the dwell times are $\tau_d = \Delta E/(P_h - P_d)$ and $\tau_a = \Delta E/(P_a - P_h)$, so $d = \tau_a/(\tau_a + \tau_d) = (P_h - P_d)/(P_a - P_d)$. Cases 1 and 2 follow from the sign of $\dot{E}$. $\square$

Remark 1. Because all dwell times have a positive lower bound, the hysteresis excludes Zeno behaviour (infinitely many switches in finite time) [3]. Average power consumed is $dP_a + (1 - d)P_d = P_h$, as energy balance requires. If a sensing condition is added to the activation guard, the lower bounds persist but the dormant dwell time has no upper bound. The original paper's three activation mechanisms (sensing, clock, emergent) then differ only in the form of the guard.

3 Interface limits on information and integration

Suppose environmental information about a variable Y reaches the computational side only through the biological sensors and an interface channel with capacity C_I bits per use. Then $Y \rightarrow X_B \rightarrow X_C \rightarrow A$ is a Markov chain, where X_B , X_C are the sensor and processed states and A is the action.

Proposition 2 (Interface bound). $I(Y; A) \leq I(X_B; X_C) \leq C_I$. For a band-limited Gaussian interface with bandwidth W and signal-to-noise ratio SNR, the capacity is $C_I = W \log_2(1 + \text{SNR})$ bits per second. By the rate-distortion theorem, the minimum expected distortion D of any decision satisfies $R(D) \leq C_I$.

Proof. The data processing inequality and the definition of channel capacity give the first claim [5]; the capacity of the Gaussian channel is the Shannon-Hartley formula [4]; the last claim combines $R(D) \leq I(Y; A)$ with the first. $\square$

This quantifies what hybridisation can and cannot add. The computational layer cannot increase the information available about Y ; it can only use that information better (lower loss). The original paper's claim of "super-additive intelligence" must therefore mean better use of the same bits, bounded by C_I .

Remark 2. In the original paper's linear model $I_{\text{HBCO}} = \alpha I_B + \beta I_C + \gamma S$, the condition $\gamma S > 0$ does not imply super-additivity. The hybrid exceeds the sum $I_B + I_C$ iff $\gamma S > (1 - \alpha)I_B + (1 - \beta)I_C$. The model is also undefined without a measurable scale for I_B , I_C , which the information bound above supplies.

The same bound limits integration across the bio-cyber boundary. For any partition-based measure that takes the boundary as the cut, such as the minimum mutual information across a bipartition, the value is at most C_I per step when the only coupling is the interface. This quantity is an integration measure, not the Φ of integrated information theory, which is defined on cause-effect repertoires of a system under perturbation [6, 7]. No claim about phenomenal consciousness follows from it, and the question remains open.

4 Learning, life, and evolution

Learning in the two substrates uses different rules (local plasticity in the biological part [8], gradient-based training in the computational part), and exact backpropagation is not biologically plausible [9]. Credit assignment across the interface must therefore be carried by signals whose total rate is bounded by C_I (Proposition 2). The original paper’s claim that HBCOs satisfy the NASA definition of life [10] is conditional: Darwinian evolution requires replication with heritable variation [11], and the model here contains no replication. Whether a specific design would satisfy the definition depends on whether it replicates, which the present framework does not assume. In contrast, the autopoietic requirement of self-production [12] is partly captured by the energy loop above, but only for the energetic aspect.

5 Safety and design requirements

Proposition 1 yields containment properties that do not depend on software.

1. **Bounded active time.** If $P_a > P_h^+$, the longest uninterrupted active period is at most $\Delta E / (P_a - P_h^+)$, independent of c . Making P_a exceed any harvest the environment can supply, and sizing ΔE accordingly, gives an energetic interlock.
2. **Controlled duty cycle.** Equation (5) shows that the fraction of time active is set by P_h and the two consumption levels, which design controls.
3. **Interface bandwidth as a capability limit.** By Proposition 2, limiting C_I limits both the information and the integration across the boundary.
4. **Process.** Hazards should be identified and scored using an established risk-management process [13], with containment and review requirements appropriate to the biological component [14]. The original paper’s probability-and-impact table had no stated scale or evidence and is removed.

Questions of moral status and legal responsibility are not settled by this model and are left to the ethics and legal literature; the measurable properties above (interface capacity, activation autonomy, replication) are the candidate inputs for any threshold.

6 What was removed

The speculative 2025–2045 timeline, the risk table, the layered-architecture and flowchart figures (replaced by the mode-and-guard model), and the intelligence and Φ formulas were removed or replaced as described. References that the original paper listed but never cited were dropped, and the mismatched citation used for the HBCO concept was not carried over.

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